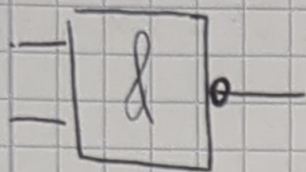


2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31



$$6.) \quad 5x - 10 = 20$$

$$\Leftrightarrow 5x = 30$$

$$\Leftrightarrow x = 6$$

Bsp. 1 "kleine" LF

$$x^2 + 5x + 6 = 0$$

$$x_{1/2} = \frac{-5}{2} \pm \sqrt{\left(\frac{5}{2}\right)^2 - 6}$$

$$= -\frac{5}{2} \pm \sqrt{\frac{25}{4} - \frac{24}{4}}$$

$$= -\frac{5}{2} \pm \sqrt{\frac{1}{4}}$$

$$= -\frac{5}{2} \pm \frac{1}{2}$$

$$\begin{aligned} x_1 &= -\frac{5}{2} + \frac{1}{2} = -2 \\ x_2 &= -\frac{5}{2} - \frac{1}{2} = -3 \end{aligned}$$

Bsp. 2 "große" LF

$$4x^2 + 6x - 4 = 0$$

$$x_{1/2} = \frac{-6 \pm \sqrt{6^2 - 4 \cdot 4 \cdot (-4)}}{2 \cdot 4}$$

$$= \frac{-6 \pm \sqrt{36 + 64}}{8}$$

$$= \frac{-6 \pm \sqrt{100}}{8}$$

$$= \frac{-6 \pm 10}{8}$$

$$x_1 = \frac{4}{8} = \frac{1}{2}$$

$$x_2 = \frac{-16}{8} = -2$$

$$4x^2 + 6x - 4 = 0 \quad | :4$$

$$\Leftrightarrow x^2 + \frac{6}{4}x - 1 = 0$$

1	1
2	2
4	3
8	4
16	5
32	6
64	7
128	8
256	9
512	10
1024	11
2048	12
4096	13
8192	14
16384	15
32768	16

$$ax^2 + bx + c = 0$$

$$p := \frac{b}{a}$$

$$q := \frac{c}{a}$$

$$\left(x + \frac{p}{2}\right)^2 = \left(x + \frac{p}{2}\right)\left(x + \frac{p}{2}\right)$$

$$= x^2 + 2 \cdot \frac{p}{2} x + \left(\frac{p}{2}\right)^2$$

$$= x^2 + p \cdot x + \left(\frac{p}{2}\right)^2$$

$$f(x) = x^2 + px + c = 0$$

$$f(x) = x^2 + px + q \stackrel{!}{=} 0 \quad | -q$$

$$\Leftrightarrow x^2 + px = -q \quad | + \left(\frac{p}{2}\right)^2 \quad \dots \text{ auf ganze Quadrate ergänzen}$$

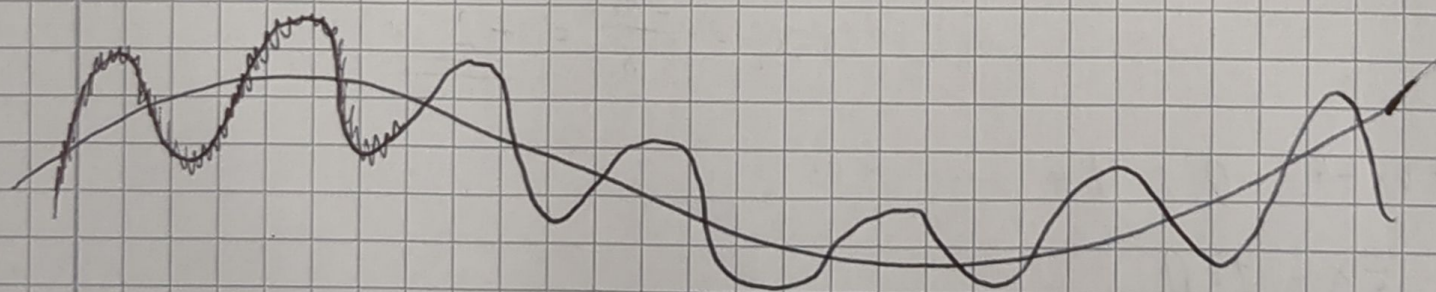
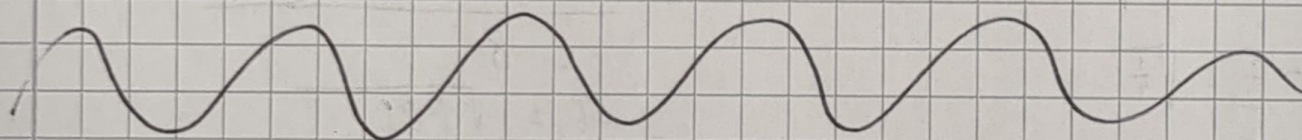
$$\Leftrightarrow x^2 + \overset{2 \cdot \frac{p}{2} x}{px} + \left(\frac{p}{2}\right)^2 = \left(\frac{p}{2}\right)^2 - q \quad | \text{ bin. Lehrsatz: } (a+b)^2 = a^2 + 2ab + b^2$$

$$\Leftrightarrow \left(x + \frac{p}{2}\right)^2 = \left(\frac{p}{2}\right)^2 - q \quad | \sqrt{\quad}$$

$$\Leftrightarrow x + \frac{p}{2} = \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}$$

$$\Leftrightarrow x = -\frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - q}$$

Herleitung pq-Formel



21.11.29

Ingo Schlapsch

1.) $81BC_{16} \xrightarrow{\text{dez}} \text{dez: } 8 \cdot 16^3 + 1 \cdot 16^2 + 11 \cdot 16^1 + \overset{12}{13} \cdot 16^0 = \underline{\underline{33212}}$

$81BC_{16} \xrightarrow{\text{bin}} \underline{\underline{1000000110111100}}_2$ ✓

$\underbrace{\hspace{2em}}_8 \quad \underbrace{\hspace{2em}}_1 \quad \underbrace{\hspace{2em}}_B \quad \underbrace{\hspace{2em}}_C$

2.) $\underbrace{1011}_B \underbrace{0011}_3 \xrightarrow{\text{hex}} \underline{\underline{B3}}$ ✓

$B_{16} \dots 1011_2$
 $3_{16} \dots 0011_2$

3.)

6	BC_{16}
8	AE_{16}
F	8B
$\underline{\underline{FGA}}$	

$\text{Anf. } C+E = \overset{12}{13}A$
 $\rightsquigarrow A+B+1 = 16$
 $6+8+1 = F$

$6AD_{16}$
 $- 36E_{16}$

 $\underline{\underline{2DF}}$ ✓

0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	A 1010
11	B 1011
12	C 1100
13	D 1101
14	E 1110
15	F 1111
16	10

$$\begin{array}{r}
 4.) \quad 1011 \cdot 101 \\
 \quad 0000 \\
 \quad \underline{1011} \\
 \quad 110111 \quad \checkmark
 \end{array}$$

$$\begin{array}{r}
 5.) \quad -4 \\
 \quad +6 \\
 \quad \hline
 \quad 2
 \end{array}$$

$$\begin{array}{r}
 4 \dots 0000 \quad 0100 \\
 \text{Flip} \quad 1111 \quad 1011 \\
 \text{Flip} + 1 \quad \dots 11 \\
 -4(2s) \quad 1111 \quad 1100 \quad \checkmark \\
 6 \quad 0000 \quad 0110 \\
 \hline
 10000 \quad 0010 \quad \dots 2 \quad \checkmark \checkmark
 \end{array}$$

$$b) \quad 4(x-3) + 2(x+1) = 20$$

$$\Leftrightarrow 4x - 12 + 2x + 2 - 20 = 0$$

$$\Leftrightarrow \overset{6}{4}x - \overset{30}{13} = 0$$

$$\Leftrightarrow \overset{6}{4}x = \overset{30}{13}$$

$$\Leftrightarrow x = \frac{13}{4} \quad 5$$

Fertig. 14:23

21. 11. 2024

$$1. a.) \quad 3x^2 = 300$$

$$\Leftrightarrow x^2 = 100$$

$$\Leftrightarrow x_{1,2} = \pm \sqrt{100}$$

$$\Rightarrow x_1 = 10$$

$$x_2 = -10$$

$$1. g.) \quad 2x^2 + 12 = 0$$

$$\Leftrightarrow x^2 + 6 = 0$$

$$\Leftrightarrow x^2 = -6$$

$$\Leftrightarrow x_{1,2} = \pm \sqrt{-6}$$

$$x_1 = -\sqrt{6}i$$

$$x_2 = \sqrt{6}i$$

$$x_{1,2} = x = \{ \}$$

$$1. h.) \quad 8x^2 - 9 = 0$$

$$\Leftrightarrow x^2 - \frac{9}{8} = 0$$

$$\Leftrightarrow x^2 = \frac{9}{8}$$

$$\Leftrightarrow x_{1,2} = \pm \sqrt{\frac{9}{8}}$$

$$x_1 = \frac{3}{\sqrt{2}}$$

$$x_2 = -\frac{3}{\sqrt{2}}$$

$$3x^2 \dots ax^2 + bx + c$$

$$3x^2 + 0x + 0 = 0 \quad | :3$$

a b c

$$\Leftrightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

p q

$$2. g.) \quad 15x^2 - 10x = 0$$

$$\Leftrightarrow 15x^2 = 10x$$

$$\Leftrightarrow 15x = 10$$

$$\Leftrightarrow x = \frac{10}{15} = \frac{2}{3}$$

$$2. h.) \quad 24x^2 = 8x$$

$$\Leftrightarrow 3x = 1$$

$$\Leftrightarrow x = \frac{1}{3}$$

$$3. g.) \quad x^2 - 4x + 7 = 0$$

$$\Rightarrow x_{1,2} = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \cdot 1 \cdot 7}}{2 \cdot 1}$$

$$= \frac{4 \pm \sqrt{16 - 28}}{2}$$

$$= \frac{4 \pm \sqrt{-12}}{2}$$

$$= \frac{4 \pm 2\sqrt{3}i}{2}$$

$$= 2 \pm \sqrt{3}i$$

$$x_1 = 2 + \sqrt{3}i$$

$$x_2 = 2 - \sqrt{3}i$$

$$3. h.) \quad x^2 + 5x + 3 = 0$$

$$\Rightarrow x_{1,2} = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 1 \cdot 3}}{2 \cdot 1}$$

$$= \frac{-5 \pm \sqrt{25 - 12}}{2}$$

$$= \frac{-5 \pm \sqrt{13}}{2}$$

$$x_1 = \frac{-5 + \sqrt{13}}{2}$$

$$x_2 = \frac{-5 - \sqrt{13}}{2}$$

$$4. g.) \quad 6x^2 - 5x - 6 = 0$$

$$\Leftrightarrow x^2 - \frac{5}{6}x - 1 = 0$$

$$\Rightarrow x_{1,2} = \frac{-(-\frac{5}{6}) \pm \sqrt{(-\frac{5}{6})^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1}$$

$$= \frac{\frac{5}{6} \pm \sqrt{\frac{25}{36} + \frac{144}{36}}}{2}$$

$$= \frac{\frac{5}{6} \pm \sqrt{\frac{169}{36}}}{2}$$

$$\Rightarrow x_1 = \frac{5}{12} + \frac{13}{12}$$

$$x_2 = \frac{5}{12} - \frac{13}{12}$$

$$3. a.) \quad x^2 + 10x + 24 = 0$$

$$\Rightarrow x_{1,2} = \frac{-10 \pm \sqrt{10^2 - 4 \cdot 1 \cdot 24}}{2 \cdot 1}$$

$$= \frac{-10 \pm \sqrt{100 - 96}}{2}$$

$$= \frac{-10 \pm \sqrt{4}}{2} = \frac{-10 \pm 2}{2}$$

$$\Rightarrow x_1 = -4 \quad x_2 = -6$$

$$x_{1,2} = \frac{-\frac{1}{5} \pm \sqrt{(\frac{1}{5})^2 - \frac{6}{25}}}{2 \cdot \frac{1}{5}}$$

$$= \frac{\frac{1}{5} \pm \sqrt{\frac{1}{25} - \frac{6}{25}}}{\frac{2}{5}}$$

$$x_{1,2} = \frac{1}{5} \pm \sqrt{\frac{1-6}{25}}$$

$$\Rightarrow x_1 = 1 + \frac{\sqrt{5}}{5}$$

$$x_2 = 1 - \frac{\sqrt{5}}{5}$$

$$5. g.) \quad 13x^2 - 5x - 6 = 12x^2$$

$$\Leftrightarrow 13x^2 - 5x - 6 + 12x^2 = 0$$

$$\Leftrightarrow 25x^2 - 5x - 6 = 0$$

$$\Leftrightarrow x^2 - \frac{1}{5}x - \frac{6}{25} = 0$$

$$4b) \quad 4x^2 - 12x + 9 = 0$$

$$\Leftrightarrow x^2 - 3x + \frac{9}{4} = 0$$

$$\begin{aligned} \Leftrightarrow x_{1/2} &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \cdot \frac{9}{4}}}{2} \\ &= \frac{3 \pm \sqrt{9 - 9}}{2} \\ &= \frac{3 \pm 0}{2} \\ &= \underline{\underline{1.5}} \end{aligned}$$

$$5a) \quad 13x^2 - 5x = 6 - 12x^2$$

$$\Leftrightarrow 25x^2 - 5x - 6 = 0$$

$$\Leftrightarrow x^2 + \frac{-5}{25}x - \frac{6}{25} = 0$$

$$\Leftrightarrow x^2 - \frac{1}{5}x - \frac{6}{25} = 0$$

$$\begin{aligned} \Rightarrow x_{1/2} &= \frac{1}{5} \pm \sqrt{\left(\frac{1}{5}\right)^2 + \frac{6}{25}} \\ &= \frac{1}{5} \pm \sqrt{\frac{1+6}{25}} = \frac{1}{5} \pm \sqrt{\frac{7}{25}} \\ &= \frac{1}{5} \pm \frac{\sqrt{7}}{5} \end{aligned}$$

~~$$6.) \quad (x+5)(x-3) + x(x-2) = (x-4)^2$$~~

Binomische Formeln

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$(a+b)(a-b) = a^2 - b^2$$

$$6.) 9.) \quad (x+5)(x-3) + x(x-2) = (x-4)^2$$

$$\Leftrightarrow x^2 - 3x + 5x - 15 + x^2 - 2x = x^2 - 4x - 4x + 16 \quad | -x^2 + 4x - 16$$

$$\Leftrightarrow 2x^2 - 15x - x^2 + 4x - 16 = 0$$

~~$$(x+5)(x-3) + x(x-2) = (x-4)^2$$~~

$$\Leftrightarrow x^2 + 4x - 31 = 0$$

$$\Rightarrow x_{1/2} = \frac{-4}{2} \pm \sqrt{\left(\frac{4}{2}\right)^2 + 31}$$

$$= -2 \pm \sqrt{4 + 31}$$

$$= -2 \pm \sqrt{35}$$

$$x_1 = -2 + \sqrt{35}$$

$$x_2 = -2 - \sqrt{35}$$

$$6a.) \quad (x-5)^2 + (2x+3)^2 = (x+1)^2 + 97$$

$$\Leftrightarrow x^2 - 10x + 25 + 4x^2 + 12x + 9 = x^2 + 2x + 1 + 97$$

$$\Leftrightarrow 4x^2 + 2x + 34 = 2x + 98$$

$$\Leftrightarrow 4x^2 = 64$$

$$\Leftrightarrow x^2 = 16$$

$$\Leftrightarrow x_{1/2} = \pm \sqrt{16}$$

$$x_1 = 4$$

$$x_2 = -4$$

Üb. Quadr. Fkt.

1.) $f(x) = -\frac{9}{5}x^2 + \frac{7}{2}x + \frac{5}{4}$

$f(x)$... höhe des Apfels in Meter

x ... Zeit in Sekunden

... Angenommen die Angabe bedeutet:

Wie hoch? $f(x) = -\frac{9}{5}x^2 + \frac{7}{2}x + \frac{5}{4}$

$f'(x) = -2\frac{9}{5}x + \frac{7}{2} = -\frac{18}{5}x + \frac{7}{2}$

$f'(x) \stackrel{!}{=} 0 = -\frac{18}{5}x + \frac{7}{2}$

$\Leftrightarrow \frac{18}{5}x = \frac{7}{2}$

$\Leftrightarrow x = \frac{5 \cdot 7}{18 \cdot 2} = \frac{35}{36} \text{ s.}$

weil $a = -\frac{9}{5}$ und $f(x) < 0$

$\rightarrow f(x)$ ist „nach unten geöffnet“

$\rightarrow$ hat bei $f'(x) = 0$ ein ~~Wert~~

maximum

$\rightarrow$ weil Polynomfkt. 2. Grades

absolutes Maximum

$f\left(\frac{35}{36}\right) = -\frac{9}{5}\left(\frac{35}{36}\right)^2 + \frac{7}{2}\left(\frac{35}{36}\right) + \frac{5}{4}$

$= \underline{\underline{2,95 \text{ m}}}$

Wie lange war d. Flugdauer?

$f(x) \stackrel{!}{=} 0 = -\frac{9}{5}x^2 + \frac{7}{2}x + \frac{5}{4}$

$\Leftrightarrow 0 = x^2 - \frac{5 \cdot 7}{9 \cdot 2}x + \frac{5 \cdot 5}{1 \cdot 4} = x^2 - \frac{35}{18}x + \frac{25}{36}$

$35^2 = 1225$

$25 \cdot 9 = 225$

~~35^2~~

$18^2 = 324$

$324 : 36 = 9$

$\rightarrow x_{1,2} = + \frac{35}{18 \cdot 2} \pm \sqrt{\left(\frac{35}{18}\right)^2 + \frac{25}{36}}$

$= \frac{35}{36} \pm \sqrt{\frac{35^2 + 25 \cdot 9}{324}}$

$= \frac{35}{36} \pm \sqrt{\frac{1225 + 225}{324}}$

$= \frac{35}{36} \pm \sqrt{\frac{1000}{18^2}}$

$= \frac{35}{36} \pm \frac{\sqrt{1000}}{18}$

$= \frac{35}{36} \pm \frac{2\sqrt{1000}}{36}$

$= \frac{35 \pm 2\sqrt{1000}}{36}$

$x_1 \approx 2,73 \text{ s}$... tats. Lösung

$x_2 \approx -0,78 \text{ s}$... theor. Lösung vor Abwurf

• Welche Abwurfhöhe

$\frac{5}{4} \text{ m}$

Üb. Quadr. Fkt.

2.) Kängurus springen 3m hoch und 10m weit

$f(x) = ?$ Höhe Nullstellen: $x_1 = 0$, $x_2 = 10$
 x ... weite

$$f(x) = ax^2 + bx + c \quad c = 0 \quad \text{weil Springer vom Boden weg}$$

$$f(x) = a(x - \overset{x_1}{0})(x - x_2) = a(x - 0)(x - 10) \\ = a(x^2 - 10x - 0x + 0) = a(x^2 - 10x) = ax^2 - 10ax$$

höchster Punkt an Scheitel $x_s = \frac{x_1 + x_2}{2} = \frac{0 + 10}{2} = 5$
ist 3

d.h. $f(5) = 3 = ax^2 - 10ax$
 $\Leftrightarrow$

d.h. $f(x=5) = 3 = a \cdot 5^2 - 10a \cdot 5$

$$\Leftrightarrow 3 = a(25 - 50)$$

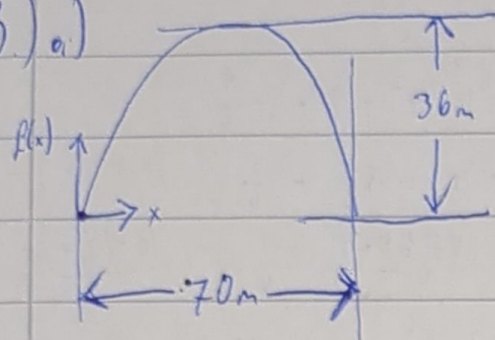
$$\Leftrightarrow 3 = a(-25)$$

$$\Leftrightarrow a = \frac{3}{-25} = -\frac{3}{25}$$

$$f(x) = -\frac{3}{25}x^2 + \frac{30}{25}x$$

Üb. Quad. Fkt.

3) a)



$$f(x) = a(x - x_1)(x - x_2)$$

$$x_1 = 0 \quad x_2 = 70$$

$$f\left(\frac{70}{2}\right) = 36$$

$$f(x) = a(x - 0)(x - 70)$$

$$= a(x^2 - 70x)$$

$$= ax^2 - 70ax$$

$$\Rightarrow b = -70a \approx 2,057$$

$$f(35) = a \cdot 35^2 - 70a \cdot 35 = 36$$

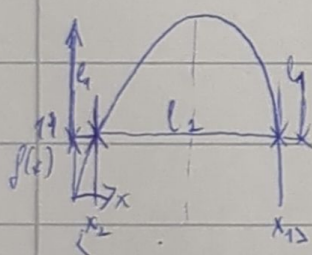
$$\Leftrightarrow a(1225 - 2450) = 36$$

$$\Leftrightarrow a(-1225) = 36$$

$$\Leftrightarrow a = -\frac{36}{1225} \approx -0,02939$$

$$\underline{\underline{f(x) \approx f(x) = -\frac{36}{1225}x^2 + \frac{72}{35}x}}$$

b) $f(x) = 19 = f(l_1) = -\frac{36}{1225}x^2 + \frac{72}{35}x$



$$\Leftrightarrow -\frac{36}{1225}x^2 + \frac{72}{35}x - 19 = 0$$

$$\Leftrightarrow x^2 + \frac{72 \cdot 1225}{-35 \cdot 36}x + \frac{19 \cdot 1225}{36} = 0$$

$$\Leftrightarrow x^2 - 70x + \frac{23275}{36} = 0$$

$$\begin{aligned} x_{1/2} &= \frac{-(-70)}{2} \pm \sqrt{\left(\frac{70}{2}\right)^2 - \frac{23275}{36}} \\ &= 35 \pm \sqrt{1225 - \frac{23275}{36}} \\ &= 35 \pm \frac{35\sqrt{17}}{6} \end{aligned}$$

$$x_1 \approx 59,05 \approx 59$$

$$x_2 \approx 10,95 \approx 11$$

$$l_2 = x_1 - x_2 \approx 59 - 11 \approx 48 \text{ m}$$

$$A = 48 \cdot 140 = \underline{\underline{6720 \text{ m}^2}}$$

Bsp. Kugelstoßer

$$f(x) = -\frac{1}{4}x^2 + 2x + \frac{9}{4}$$

max Höhe bei $f'(x) = 0$

$$f'(x) = -2\frac{1}{4}x + 2 \stackrel{!}{=} 0$$

$$\Leftrightarrow -\frac{1}{2}x + 2 = 0$$

$$\Leftrightarrow \frac{1}{2}x = 2$$

$$\Leftrightarrow x = 4$$

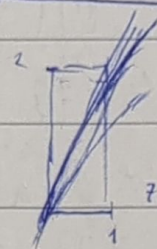
$$f'(0) = 0 + 2 = 2$$

$$f(x=4) = -\frac{1}{4} \cdot 4^2 + 2 \cdot 4 + \frac{9}{4}$$

$$= -4 + 8 + \frac{9}{4}$$

$$= \frac{28 \cdot 4}{4} + \frac{9}{4} = \frac{112+9}{4} = \frac{121}{4} = 30,25$$

max Flughöhe $\approx 30,25 \text{ m}$



zu steile Parabel für eine gute Weite

Weite d. Wurfs

$$f(x) \stackrel{!}{=} 0 = -\frac{1}{4}x^2 + 2x + \frac{9}{4}$$

l. $\left(-\frac{1}{4}\right)$

Weite $= 9 \text{ m}$

$$\Leftrightarrow 0 = x^2 - 8x - 9$$

$$\Rightarrow x_{1,2} = -\frac{-8 \pm \sqrt{(-8)^2 + 4 \cdot 9}}{2}$$

$$= 4 \pm \sqrt{16 + 9}$$

$$= 4 \pm \sqrt{25}$$

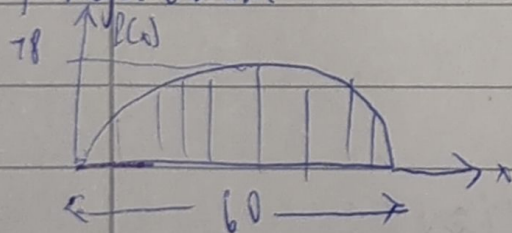
$$= 4 \pm 5$$

$x_1 = 9 \text{ m}$... gesuchte Lösung

$x_2 = -1$... nicht d. ges. Lösung

Abwurfhöhe: $\frac{9}{4} = 2,25 \text{ m}$

Bsp. Tragseilbrücke



Grundlagen quadratische Gleichungen / ermitteln Sie die jeweilige Lösungsmenge

Bsp. 1)

$$(x+1)^2 = 29 - x$$

$$\Leftrightarrow x^2 + 2x + 1 - 29 + x = 0$$

$$\Leftrightarrow x^2 + 3x - 28 = 0$$

$$\Rightarrow x_{1,2} = -\frac{3}{2} \pm \sqrt{\left(\frac{3}{2}\right)^2 + 28}$$

$$= -\frac{3}{2} \pm \sqrt{\frac{9}{4} + \frac{112}{4}}$$

$$= -\frac{3}{2} \pm \frac{\sqrt{121}}{2}$$

$$= -\frac{3}{2} \pm \frac{11}{2}$$

$$\Rightarrow x_1 = -\frac{3}{2} + \frac{11}{2} = \frac{8}{2} = \underline{4}$$

$$x_2 = -\frac{3}{2} - \frac{11}{2} = \frac{-14}{2} = \underline{\underline{-7}}$$

$$\underline{\underline{x \in \{4, -7\}}}$$

Bsp 2:

$$(x-2)(x+2) = 4x + 1$$

$$\Leftrightarrow x^2 - 4 - 4x - 1 = 0$$

$$\Leftrightarrow x^2 - 4x - 5 = 0$$

$$\Rightarrow x_{1,2} = -\frac{-4}{2} \pm \sqrt{\left(\frac{4}{2}\right)^2 + 5}$$

$$= \frac{4}{2} \pm \sqrt{4+5}$$

$$= 2 \pm \sqrt{9}$$

$$= 2 \pm 3$$

$$\Rightarrow x_1 = 2+3 = \underline{5}$$

$$x_2 = 2-3 = \underline{\underline{-1}}$$

$$\underline{\underline{x \in \{-1, 5\}}}$$

MAA 28.11.24

Diskriminante

$$x_{1,2} = -\frac{b}{2} \pm \sqrt{\left(\frac{b}{2}\right)^2 - a}$$

bei Mitternachts/PQ-Formel

Wert unter der Wurzel lässt schließen auf Anzahl d. Lösungen

$D > 0$... 2 Lösungen

$D = 0$... 1 (doppelte) Lösung

$D < 0$... keine (reellen) Lösungen

$$x^2 - 2x - 3 = 0 \quad -\frac{-2}{2} \pm \sqrt{\left(\frac{-2}{2}\right)^2 + 3}$$

$1+3=4 > 0 \rightarrow 2 \text{ Lsg.}$

a.) $x + y = 10$

$x \cdot y = 24$

$\Rightarrow y = 10 - x$

$\Rightarrow x \cdot (10 - x) = 24$

$\Leftrightarrow 10x - x^2 - 24 = 0$

$\Leftrightarrow x^2 - 10x + 24 = 0$

$x = 6 : y = 10 - 6 = 4$

$x = 4 : y = 10 - 4 = 6$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^2 = (a+b)(a+b)$$

$$= a^2 + ab + ba + b^2 = a^2 + 2ab + b^2$$

$$(x+1)(x+1) = x^2 + x + x + 1$$

$$x_{1,2} = -\frac{-10}{2} \pm \sqrt{\left(\frac{-10}{2}\right)^2 - 24}$$

$$= 5 \pm \sqrt{25 - 24}$$

$$= 5 \pm 1$$

$x \in \{6, 4\}$

$\mathbb{L} = \{ \{6, 4\}, \{4, 6\} \}$

$\{ \{6^x, 4^x\}, \{4^x, 6^x\} \}$

$$T9 \text{ I: } a + b = 10$$

$$\text{II: } a \cdot b = 24$$

$$\begin{aligned} \stackrel{(2)}{\Rightarrow} a_{1,2} &= -\frac{-10}{2} \pm \sqrt{\left(\frac{10}{2}\right)^2 - 24} \\ &= 5 \pm \sqrt{5^2 - 24} \end{aligned}$$

$$\begin{aligned} &\sqrt{\left(\frac{10}{2}\right)^2 - 40} \\ \Leftrightarrow &\sqrt{\frac{100}{4} - \frac{4 \cdot 40}{4}} \\ \Leftrightarrow &\sqrt{\frac{160}{4}} \Leftrightarrow \frac{\sqrt{160}}{\sqrt{4}} \\ \Leftrightarrow &\frac{13}{2} \end{aligned}$$

$$= 5 \pm \sqrt{25 - 24}$$

$$= 5 \pm \sqrt{1}$$

$$= 5 \pm 1$$

$$\Rightarrow a_1 = 6$$

$$a_2 = 4$$

$$\text{I} \Rightarrow b \stackrel{(1)}{=} 10 - a$$

$$\text{II} \stackrel{(1)}{\Rightarrow} a \cdot (10 - a) = 24$$

$$\Leftrightarrow 10a - a^2 - 24 = 0$$

$$\Leftrightarrow a^2 - 10a + 24 \stackrel{(2)}{=} 0$$

$$\stackrel{(1)}{\Rightarrow} b_1 = 10 - a_1 = 10 - 6 = 4$$

$$b_2 = 10 - a_2 = 10 - 4 = 6$$

$$\Rightarrow L_1 = \{\{x|y\} | \{6|4\}, \{4|6\}\}$$

$$\begin{array}{r} | x = 4 \\ \hline x - 2 = 6 \quad | +2 \\ \hline x = 8 \\ \hline x - y = 4 \end{array}$$

$$0 = 10y - y^2 - 24$$

$$\Leftrightarrow -y^2 + 10y - 24 = 0 \quad | :(-1)$$

$$\Leftrightarrow y^2 - 10y + 24 = 0$$

$$\begin{array}{ll} T10 \text{ I: Diff. zw. Zahlen ist } 3 & \dots \quad \overset{v_1}{x - y = 3} \quad \text{oder} \quad \overset{v_2}{y - x = 3} \\ \text{II: Prod.} \quad -11 \quad \dots \quad 40 & \dots \quad x \cdot y = 40 \end{array}$$

$$\begin{aligned} v_1: \text{I} \Rightarrow x - y = 3 &\Rightarrow x \stackrel{(1)}{=} 3 + y \quad \text{II} \stackrel{(1)}{\Rightarrow} (3 + y) \cdot y = 40 \Leftrightarrow 3y + y^2 = 40 \\ &\Rightarrow \end{aligned}$$

$$576 - 512$$

$$\sqrt{-24^2} = \sqrt{-576}$$

$$x = \sqrt{4}$$

$$\Rightarrow x = 2 \quad \text{weil } 2^2 = 4$$

$$T10 \text{ Diff. zw. Zahlen gleich } 3$$

$$\dots \quad |a - b| = 3$$

$$x = \sqrt{-4}$$

$$\Leftrightarrow a - b = 3 \quad \vee \quad b - a = 3$$

MAM 02.12.24

$$x = x^2 + x + 1$$

(12)

19.) $U = 70 \text{ cm}$

$A = 300 \text{ cm}^2$

Rechteck

I $a \cdot b = 300 \text{ cm}^2$

II $2(a+b) = 70 \text{ cm}$

I $\Rightarrow a = \frac{300}{b}$

II $\Rightarrow 2\left(\frac{300}{b} + b\right) = 70 \text{ cm} \quad | :2$

$\Leftrightarrow \frac{300 + b^2}{b} = 35 \text{ cm} \quad | \cdot$

$\Leftrightarrow 300 + b^2 = 35b \quad | \cdot$

$\Leftrightarrow b = \frac{35 \text{ cm}}{300 \text{ cm}^2}$

$\Leftrightarrow a = \frac{300}{35} = \frac{600}{7}$

II: $2(a+b) = 70$

$\Rightarrow a+b = 35$

$\Leftrightarrow a = 35 - b$

(2) $\Rightarrow (35-b) \cdot b = 300$

$\Leftrightarrow 35b - b^2 = 300$

$\Leftrightarrow -b^2 + 35b - 300 = 0$

$\Leftrightarrow b^2 - 35b + 300 = 0$

$$b_{1/2} = \frac{-(-35) \pm \sqrt{(-35)^2 - 4 \cdot 1 \cdot 300}}{2 \cdot 1}$$

$$= \frac{35 \pm \sqrt{1225 - 1200}}{2}$$

$$= \frac{35 \pm \sqrt{25}}{2}$$

$$= \frac{35 \pm 5}{2}$$

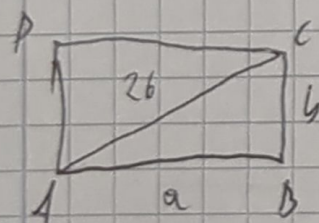
$\Rightarrow b_1 = \frac{40}{2} = 20$

$b_2 = \frac{30}{2} = 15$

$a_1 = 35 - b_1 = 35 - 20 = 15$

$a_2 = 35 - b_2 = 35 - 15 = 20$

20.) $U = 68$



I: $2(a+b) = 68$

II: $a^2 + b^2 = 26^2$

I: $a+b = 34$

$\Leftrightarrow a = 34 - b$

II: $(34-b)^2 + b^2 = 26^2$

$\Leftrightarrow 34^2 - 68b + b^2 + b^2 = 26^2$

$\Leftrightarrow 2b^2 - 68b + 34^2 - 26^2 = 0$

$\Leftrightarrow b^2 - 34b + 240 = 0$

$\Rightarrow b_{1/2} = \frac{-(-34) \pm \sqrt{(-34)^2 - 4 \cdot 1 \cdot 240}}{2 \cdot 1}$

$$= \frac{34 \pm \sqrt{1156 - 960}}{2}$$

$$= \frac{34 \pm \sqrt{196}}{2}$$

$$= \frac{34 \pm 14}{2}$$

$\Rightarrow b_1 = \frac{34+14}{2} = 24$

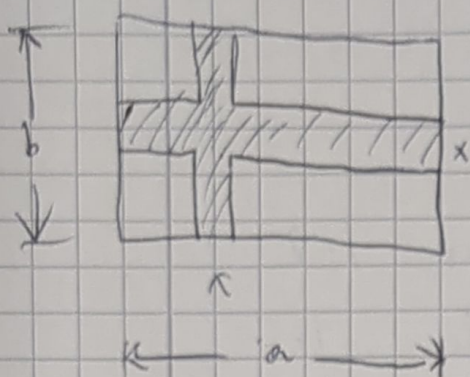
$\Rightarrow b_2 = \frac{34-14}{2} = 10$

$a_1 = 34 - b_1 = 34 - 24 = 10$

$a_2 = 34 - b_2 = 34 - 10 = 24$

~~11. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100.~~

21.) Flayge



Wie groß muss ~~100~~ x sein, sodass der Flächeninhalt d. Kreuzes = Flächeninh. der restl. Flayge

$$A_K = \cancel{a \cdot x} (a \cdot x) + (x \cdot (b - x)) = ax + bx - x^2$$

$$A_F = (a - x)(b - x) = ab - ax - bx + x^2$$

$$A_F = A_K \Rightarrow -x^2 + ax + bx = x^2 + (-a - b)x + ab$$

$$\Leftrightarrow -2x^2 + 2ax + 2bx - ab = 0$$

$$\Leftrightarrow x^2 - ax - bx + ab = 0$$

$$\Leftrightarrow x^2 - (a+b)x + ab = 0$$

$$\Rightarrow x_{1/2} = -\frac{-(a+b)}{2} \pm \sqrt{\left(\frac{a+b}{2}\right)^2 - ab}$$

$$x = \frac{200}{2} \pm \sqrt{\frac{40000}{4} - 9600}$$

$$= 100 \pm \sqrt{10000 - 9600}$$

$$= 100 \pm \sqrt{400}$$

$$= 100 \pm 20$$

$$x_1 = 120 \text{ cm}$$

$$x_2 = 80 \text{ cm}$$

$$a+b=$$

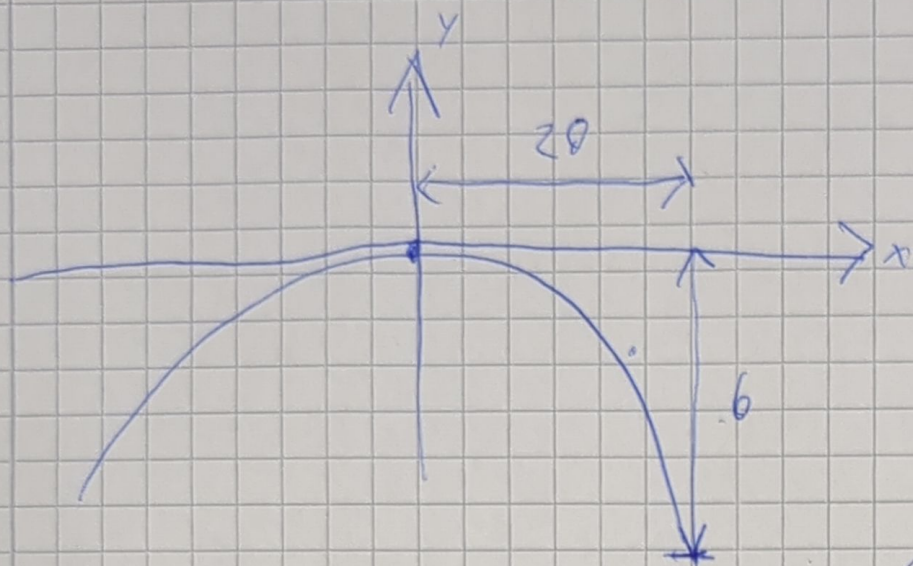
$$a = 120$$

$$b = 80$$

$$a+b = 200$$

$$ab = 9600$$

Mat 05.12.



$f(x) = a x^2$ ~~ist. f. g.~~ ist. f. g.
 $f(6) = -20$
 $f(0) = 0$

$\Rightarrow f(6) = a \cdot 6^2 = -20$
 $\Rightarrow a \cdot 36 = -20$
 $\Rightarrow a = \frac{-20}{36}$

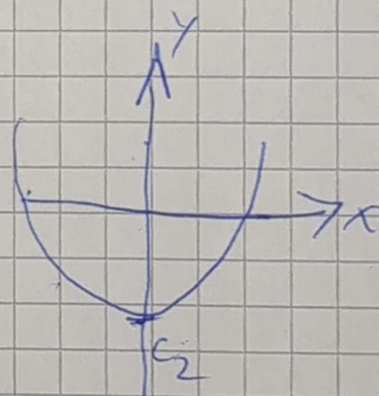
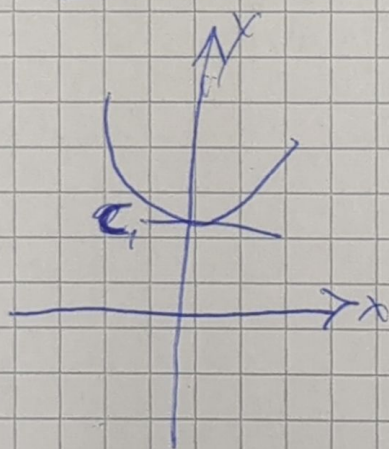
$\Rightarrow f(20) = 6 = a \cdot 20^2$

$\Rightarrow \frac{6}{20^2} = a = \underline{\underline{0,015 \text{ m}^{-1}}}$

$y = a x^2 + c$

$c_1 > 0$

$c_2 < 0$



$f(x) = (x - x_1)(x - x_2) = x^2 - x \cdot x_2 - x x_1 + x_1 x_2$

Klasse: 3a APC

Name: Ingo Schlöpsky

Datum: 16.12.2024

Punkte: 2

- 1) Grundlagen zu einfachen Gleichungen: Bestimmen Sie die jeweilige Lösung!

2P./

2

$$\begin{aligned}
 18 - 5(x + 2) &= 3 && | -18 \\
 \Leftrightarrow -5x - 10 &= -15 && | +10 \\
 \Leftrightarrow -5x &= -5 \\
 \Leftrightarrow x &= 1
 \end{aligned}$$

- 2) Grundlagen zu quadratischen Gleichungen: Bestimmen Sie die jeweilige Lösung!

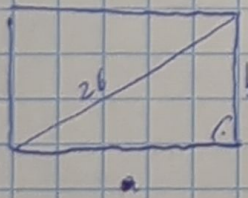
2P./

2

$$\begin{aligned}
 x^2 + 10x + 24 &= 0 && \text{mit } 5^2 = 25 \\
 x_{1,2} &= -\frac{10}{2} \pm \sqrt{\left(\frac{10}{2}\right)^2 - 24} \\
 &= -5 \pm \sqrt{5^2 - 24} \\
 &= -5 \pm \sqrt{25 - 24} \\
 &= -5 \pm 1 \\
 x_1 &= -5 + 1 = -4 \\
 x_2 &= -5 - 1 = -6
 \end{aligned}$$

- 3) Grundlagen zu quadratischen Gleichungen – Textbeispiel: Bestimmen Sie die jeweilige Lösung! 2P./

Der Umfang eines Rechteckes beträgt 68cm – die Diagonale ist 26cm lang.
Bestimmen Sie die Seitenlängen und die Fläche.



$$\begin{aligned}
 \text{I: } a + b &= \frac{68}{2} \\
 \Leftrightarrow a &= 34 - b
 \end{aligned}$$

$$\text{II: } a(34 - b)^2 + b^2 = 26^2$$

$$\Leftrightarrow 34^2 - 2 \cdot 34b + b^2 + b^2 = 26^2$$

$$\Leftrightarrow 2b^2 - 68b + 480 = 0$$

$$\Leftrightarrow b^2 - 34b + 240 = 0$$

$$\begin{aligned}
 \text{I: } 2(a+b) &= 68 \\
 \text{II: } a^2 + b^2 &= 26^2
 \end{aligned}$$

$$\begin{aligned}
 b_{1,2} &= -\frac{-34}{2} \pm \sqrt{\left(\frac{-34}{2}\right)^2 - 240} \\
 &= 17 \pm \sqrt{289 - 240} \\
 &= 17 \pm \sqrt{49} \\
 &= 17 \pm 7
 \end{aligned}$$

$$b_1 = 17 + 7 = 24$$

$$b_2 = 17 - 7 = 10$$

$$\begin{aligned}
 \frac{34}{2} &= 17 && 34^2 = 1156 \\
 17^2 &= 289 && 26^2 = 676 \\
 289 - 240 &= 49 && 34^2 - 26^2 = 480
 \end{aligned}$$

$$\begin{aligned}
 17^2 &= 289 \\
 \sqrt{49} &= 7
 \end{aligned}$$

$$a_1 = 34 - b_1 = 34 - 24 = 10$$

$$a_2 = 34 - b_2 = 34 - 10 = 24$$

irgendwas verrechnet?
Ergebnis ist falsch!

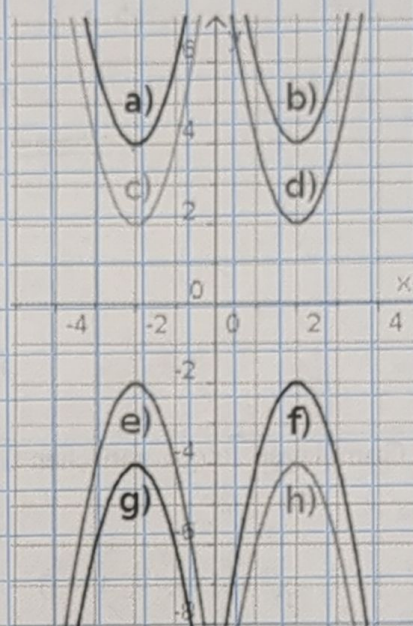
$$\begin{aligned}
 L_1 &= \{10, 24\} \\
 L_2 &= \{24, 10\}
 \end{aligned}$$

4) Grundlagen zu quadratischen Funktionen: Bestimmen Sie die jeweilige Lösung!

2P./ 2

Neu

a	$y = 2(x + 2)^2 + 4$	---	a) - Parabel
c	$y = 2(x + 2)^2 + 2$	---	b) - Parabel
d	$y = 2(x - 2)^2 + 2$	---	c) - Parabel
b	$y = 2(x - 2)^2 + 4$	---	d) - Parabel
l	$y = -2(x - 2)^2 - 2$	---	e) - Parabel
h	$y = -2(x - 2)^2 - 4$	---	f) - Parabel
e	$y = -2(x + 2)^2 - 2$	---	g) - Parabel
g	$y = -2(x + 2)^2 - 4$	---	h) - Parabel



Auswertung

richtig: 0 | falsch: 0

$$\text{Bsp } 7^{x+2} = 97$$

$$\Leftrightarrow \log_7(7^{x+2}) = \log_7(97)$$

$$\Leftrightarrow (x+2) \log_7(7) = \log_7(97)$$

$$\Leftrightarrow x+2 = \log_7(97)$$

$$\Leftrightarrow x = \log_7(97) - 2$$

$$\approx 2,35 - 2 \approx 0,35$$

OP.

ist sich nicht mehr
ausgegangen, wegen
Fehlersuche bei Bsp. 3